

MATH-3108-103-70982-2023FS-Linear Algebra I Exam 2

Michael Colletti

TOTAL POINTS

92 / 100

QUESTION 1

1 Question 1 3 / 3

✓ - 0 pts Correct

QUESTION 2

2 Question 2 3 / 3

✓ - 0 pts Correct

QUESTION 3

3 Question 3 3 / 3

✓ - 0 pts Correct

QUESTION 4

4 Question 4 0 / 3

✓ - 3 pts Incorrect. The correct answer is (i).

QUESTION 5

5 Question 5 4 / 4

✓ - 0 pts Everyone gets credit for this problem. Not enough information was given, which made a relatively straightforward problem far more challenging.

Note that, as stated, the correct answer to this problem would have been to leave all four boxes blank.

If you want a sample problem to study for the final, add the assumption that A is a 4×4 matrix.

QUESTION 6

6 Question 6 2 / 4

✓ - 2 pts One correct answer, no incorrect answers.

The fully correct answer is (ii) and (iii).

QUESTION 7

7 Question 7 5 / 6

✓ - 1 pts Did not justify how scalar multiplication generates a vector not in H

QUESTION 8

Question 8 8 pts

8.1 Question 8a 6 / 6

✓ - 0 pts Correct

8.2 Question 8b 0 / 2

✓ - 2 pts Incorrect. Correct answer is ii).

QUESTION 9

Question 9 14 pts

9.1 Question 9a 4 / 4

✓ - 0 pts Correct

9.2 Question 9b 10 / 10

✓ - 0 pts Correct

QUESTION 10

10 Question 10 10 / 10

✓ - 0 pts Correct

QUESTION 11

11 Question 11 8 / 8

✓ - 0 pts Correct

QUESTION 12

12 Question 12 12 / 12

✓ - 0 pts Correct

QUESTION 13

13 Question 13 6 / 6

✓ - 0 pts Correct

QUESTION 14

Question 14 10 pts

14.1 Question 14a 6 / 6

✓ + 6 pts Correct

14.2 Question 14b 4 / 4

✓ - 0 pts Correct

QUESTION 15

15 Question 15 6 / 6

✓ + 6 pts Correct

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Math 3108 – Linear Algebra
Exam 2
Prof. Runnion
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Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has seven sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO

Name: Michael CollettiStudent ID: 18217484**Multiple Choice and True/False.** Fill in the box corresponding to the correct answer.

1. (3 pts) A basis for a vector space V is a _____ of vectors in V .
- ☐ i) minimal spanning set ☐ ii) maximal linearly independent set
- ☒ iii) both (a) and (b) ☐ iv) neither (a) nor (b)
2. (3 pts) For an $n \times n$ matrix A , if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, then A is row equivalent to the identity matrix.
- ☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true
3. (3 pts) If A is an invertible $n \times n$ matrix, then the columns of A^{-1} are linearly independent.
- ☒ i) Always true ☐ ii) Sometimes true ☐ iii) Never true
4. (3 pts) If A and B are invertible $n \times n$ matrices, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$
- ☐ i) Always true ☒ ii) Sometimes true ☐ iii) Never true
5. (4 pts) If the third row of A is the sum of four times the first row and five times the second row, which of the following is true? Select all that apply.
- ☐ i) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- ☒ ii) The column space of A is a subspace of $\mathbb{R}^4$.
- ☐ iii) The rows of A span $\mathbb{R}^4$.
- ☒ iv) The determinant of A is zero.
6. (4 pts) Let A be an $m \times n$ matrix. Which of the following is a subspace of $\mathbb{R}^n$? Select all that apply.
- ☐ i) $\text{Col } A$ (the column space of A)
- ☒ ii) $\text{Nul } A$ (the null space of A)
- ☐ iii) $\text{Row } A$ (the row space of A)

$$\begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

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Open Answer. Show ALL your work! Place your final answer in the box provided, if applicable.

7. (6 points) Show that the set $H = \left\{ \begin{bmatrix} a+2b \\ 3b+4 \\ a-b \end{bmatrix} : a, b \text{ real} \right\}$ is *not* closed with respect to scalar multiplication.

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} a + \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} b + \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}$$

Not closed to scalar multiplication
because of a constant 4 in the
set H

$$cH \neq ca + cb$$

8. a) (6 points) Find a matrix A such that $W = \left\{ \begin{bmatrix} c-6d \\ d \\ c \end{bmatrix} : c, d \text{ real} \right\} = \text{Col } A$.

$$\begin{bmatrix} 1 & -6 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Solution: $A = \begin{bmatrix} 1 & -6 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

- b) (2 points) The set W is a subspace of. (Fill in the box.)

☒ i) $\mathbb{R}^2$ ☐ ii) $\mathbb{R}^3$ ☐ iii) Neither of these.

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9. The matrices A and B given below are row equivalent. (You do *not* need to check this.)

$$A = \begin{bmatrix} 2 & -4 & -1 & -3 & 5 & 2 \\ -1 & 2 & -4 & -3 & -7 & -1 \\ -1 & 2 & -13 & -12 & -16 & -1 \\ 5 & -10 & 5 & -10 & 20 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- a) (4 points) Find a basis for the column space of A .

$$\text{span} \left\{ \begin{bmatrix} 2 \\ -1 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ -4 \\ -13 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ -3 \\ -12 \\ -10 \end{bmatrix} \right\} = \text{basis col } A$$

- b) (10 points) Find a basis for the null space of A .

$$x_1 - 2x_2 + 3x_5 + x_6 = 0 \Rightarrow x_1 = 2x_2 - 3x_5 - x_6$$

$$x_3 + x_5 = 0 \Rightarrow x_3 = -x_5$$

$$x_4 = 0$$

x_2, x_5, x_6 are free

$$\begin{bmatrix} 2 \\ -1 \\ -1 \\ 5 \end{bmatrix} x_2 + \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \end{bmatrix} x_5 + \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} x_6$$

$$\text{Span} \left\{ \begin{bmatrix} 2 \\ -1 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} = \text{basis Nul } A$$

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10. (10 points) Consider the system of equations
$$\begin{cases} x + 2z = 3 \\ -2x + y + z = 4 \\ x - 3y + z = 1 \end{cases}$$

Given that $\begin{vmatrix} 1 & 0 & 2 \\ -2 & 1 & 1 \\ 1 & -3 & 1 \end{vmatrix} = 14$, use Cramer's Rule to solve for the variable z only.

$$\frac{\det(b)}{\det A} = x_i$$

$$\begin{vmatrix} 1 & 0 & 3 \\ -2 & 1 & 4 \\ 1 & -3 & 1 \end{vmatrix}$$

$$\begin{aligned} & 1(1+12) + 3(6-1) \\ & 13 + 15 \\ & 28 \end{aligned}$$

$$\Rightarrow \frac{28}{14} = 2$$

Solution: $z =$

2

11. (8 points) Assume that the matrices shown below are partitioned conformably for block multiplication. Compute the product.

$$\begin{bmatrix} I & -E \\ 0 & I \end{bmatrix} \begin{bmatrix} W & X \\ Y & Z \end{bmatrix}$$

$$\begin{bmatrix} IW - EY & IX - EZ \\ 0W + IY & 0X + IZ \end{bmatrix} \Rightarrow \begin{bmatrix} W - EY & X - EZ \\ Y & Z \end{bmatrix}$$

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12. (12 points) Find the LU factorization of the matrix $A = \begin{bmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{bmatrix}$.

 $U = \text{echelon form}$

$$\begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 3 & -3 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$L = \frac{1}{2} \begin{bmatrix} 2 \\ -6 \\ 8 \end{bmatrix}, \frac{1}{3} \begin{bmatrix} -3 \\ 3 \end{bmatrix}, [1]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{bmatrix}$$

Solution: $L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

13. (6 points) Show that the set of all polynomials of the form $p(t) = a + bt^2$ where a and b are real is closed with respect to vector addition.

if $b = x + y$

& $a = q + p$

& $x, y, q, \& p$ are
real

$(q+p) + (x+y)t^2$ still has the form
 $a + bt^2$, thus $p(t)$ is closed with
vector addition

$$(q + xt^2) + (p + yt^2) \Rightarrow (q+p) + (x+y)t^2$$

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14. a) (6 points) Use the method of your choice to calculate A^{-1} given $A = \begin{bmatrix} -2 & -3 \\ 2 & 1 \end{bmatrix}$.

$$\frac{1}{\det A} \text{adj} A = A^{-1}$$

$$\det A = -2 - (-6) = 4$$

$$\frac{1}{4} \begin{bmatrix} 1 & 3 \\ -2 & -2 \end{bmatrix}$$

$$\frac{1}{4} \begin{bmatrix} 3 \\ -1/2 \end{bmatrix}$$

$$\begin{aligned} -1/2 + 3/2 &= 1 \\ 1/2 - 1/2 &= 0 \end{aligned} \quad \begin{aligned} -3/2 + 3/2 &= 0 \\ 3/2 - 1/2 &= 1 \end{aligned}$$

$$\text{Solution: } A^{-1} = \begin{bmatrix} 1/4 & 3/4 \\ -1/2 & -1/2 \end{bmatrix}$$

- b) (4 points) Use the inverse you found in part (a) to solve $A\mathbf{x} = \mathbf{b}$ given $\mathbf{b} = \begin{bmatrix} -8 \\ 4 \end{bmatrix}$.

$$\mathbf{x} = A^{-1} \mathbf{b}$$

$$\begin{bmatrix} 1/4 & 3/4 \\ -1/2 & -1/2 \end{bmatrix} \begin{bmatrix} -8 \\ 4 \end{bmatrix}$$

$$-8(1/4) + (3/4)(4)$$

$$-8(-1/2) + 4(-1/2)$$

$$\begin{bmatrix} -2 + 3 \\ 4 - 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\text{Solution: } \mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

15. (6 points) Find the area of the parallelogram whose vertices are $(0, -2)$, $(5, -2)$, $(-3, 1)$, and $(2, 1)$. Place your answer in the box provided.

subtract $(0, -2)$ from all

$$(0, 0), (5, 0), (-3, 3), (2, 3)$$

$$\begin{bmatrix} 5 & 0 \\ -3 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -3 \\ 0 & 3 \end{bmatrix}$$

$$\text{Area} = 15$$

$$15 - 0$$

